Lecture 25 Scribe

L25_Problem_Solving__Tutorial (1).pdf
PDF

Act as a professional Lecture Scribe for CSE 400: Fundamentals of Probability in Computing and generate a comprehensive, exam-ready lecture scribe for Lecture 25: Problem Solving and Tutorial Session based strictly on the provided context in L25_Problem_Solving__Tutorial (1).pdf. In accordance with the mandatory Chain-of-Thought (CoT) instructions, you must reason step-by-step through every problem presented, including the identification of the random variable type (such as Rayleigh or Gaussian), the selection of the corresponding PDF or CDF, and the logical execution of the solution. Ensure these reasoning steps are explicitly included in the final document to allow students to follow the logical flow during revision, specifically explaining why a particular bound—like Markov's or Chebyshev's—is selected for a given problem. The scribe must begin with the objective statement: "Objective: This scribe serves as an academic reference for exam preparation for Lecture 25". All content and logical derivations must be traceable only to the provided tutorial materials, without introducing outside intuition or external examples. Reason through the calculation of power in decibels (dB), the transformation of variables, and the interpretation of tail probabilities, making the logical dependencies between these concepts explicit. Ensure all mathematical steps, including integrals and the standard normal conversion, make it while while maintaining a formal, academic, and instructional tone throughout.

Objective: This scribe serves as an academic reference for exam preparation for Lecture 25

CSE 400: Fundamentals of Probability in Computing

Lecture 25: Problem Solving and Tutorial Session (Inequalities)

This lecture focuses on applying **probabilistic inequalities**—primarily **Markov, Chebyshev, Jensen, and Chernoff bounds**—to solve structured problems. Each solution explicitly follows a **step-by-step reasoning process**, including identification of the random variable, choice of inequality, and justification.

Q1: Markov's Inequality (Income Problem)

Given:

- Random variable X: Annual income
- $\mathrm{E}[X] = 40{,}000$

(a) Find $P(X \geqslant 120{,}000)$

Step 1: Identify inequality

Since:

- $X \geqslant 0$
- Only mean is known

➡ Use **Markov's Inequality**:

$$P(X \geqslant a) \leqslant \frac{\mathrm{E}[X]}{a}$$

Step 2: Apply

$$P(X \geqslant 120{,}000) \leqslant \frac{40{,}000}{120{,}000} = \frac{1}{3}$$

Final Answer:

$$P(X \geqslant 120{,}000) \leqslant \frac{1}{3}$$

(b) Can we bound $P(X < 10{,}000)$?

Reasoning:

- Markov applies only to $P(X \geqslant a)$
- This is a **lower-tail probability**

Conclusion:

❌ **Cannot use Markov's inequality**
➡ Because it only bounds **upper tail probabilities**

Q2: Chebyshev's Inequality (Resistor Problem)

Given:

- Mean $\mu = 100$
- Standard deviation $\sigma = 2$
- Defective if outside $[94, 106]$

(a) Probability of defect

Step 1: Convert interval to deviation

$$|X - 100| \geqslant 6$$

$$k = \frac{6}{2} = 3$$

Step 2: Apply Chebyshev

$$P(|X - \mu| \geqslant k\sigma) \leqslant \frac{1}{k^2}$$

$$P(\text{defective}) \leqslant \frac{1}{3^2} = \frac{1}{9}$$

(b) Expected defective out of 10,000

$$\text{Max expected} = 10{,}000 \times \frac{1}{9} \approx 1111$$

Q3: Convexity and Expectation Bound

Given:

- $0 \leqslant Y \leqslant 1$
- $\mathrm{E}[Y] = \mu$

Step 1: Identify tool

- Function $g(x) = e^x$ is **convex**
  ➡ Use **Jensen's inequality idea via convex combination**

Step 2: Express Y as mixture

Since $Y \in [0, 1]$:

$$Y = (1 - Y) \cdot 0 + Y \cdot 1$$

Step 3: Apply convexity

$$e^{tY} \leqslant (1 - Y)e^0 + Y e^t$$

$$e^{tY} \leqslant (1 - Y) + Y e^t$$

Step 4: Take expectation

$$\mathrm{E}[e^{tY}] \leqslant 1 - \mu + \mu e^t$$

Q4: Jensen's Inequality (Power Function)

Given:

- $X > 0$
- $C(X) = X^a, \ a > 1$

Step 1: Identify convexity

Function $f(x) = x^a$ is **convex for** $a > 1$

Step 2: Apply Jensen

$$\mathrm{E}[X^a] \geqslant (\mathrm{E}[X])^a$$

Q5: Chebyshev General Form

To Show:

$$P(|X-\mu| \geqslant k\sigma) \leqslant \frac{1}{k^2}$$

Derivation:

Apply Markov on:

$$(X-\mu)^2$$

$$P((X-\mu)^2 \geqslant k^2\sigma^2) \leqslant \frac{\mathrm{E}[(X-\mu)^2]}{k^2\sigma^2}$$

$$= \frac{\sigma^2}{k^2\sigma^2} = \frac{1}{k^2}$$

Q6: Production Problem

(a) $P(X \geqslant 1000)$

Given:

- Mean = 500

➡ Use **Markov**:

$$P(X \geqslant 1000) \leqslant \frac{500}{1000} = \frac{1}{2}$$

(b) $P(400 \leqslant X \leqslant 600)$

Step 1: Convert

$$|X-500| < 100$$

$$k = \frac{100}{\sqrt{100}} = 10$$

Step 2: Chebyshev

$$P(|X-500| \geqslant 100) \leqslant \frac{1}{100}$$

Step 3: Complement

$$P(400 \leqslant X \leqslant 600) \geqslant 1 - \frac{1}{100} = 0.99$$

Q7: Chernoff Bound Setup

Given:

- $X_i = \pm 1$, equal probability

(a) Compute $\mathrm{E}[e^{tX_i}]$

$$\mathrm{E}[e^{tX_i}] = \frac{1}{2}(e^t + e^{-t})$$

Use inequality:

$$\frac{e^t + e^{-t}}{2} \leqslant e^{t^2/2}$$

(b) For sum S_n

Independence ⇒ multiply MGFs:

$$\mathrm{E}[e^{tS_n}] = \prod \mathrm{E}[e^{tX_i}] \leqslant e^{nt^2/2}$$

Q8: Pairing Problem

Goal:

Bound probability that $\leqslant 30$ mixed pairs

Approach:

- Let X = number of mixed pairs
- Use **Chernoff bound (lower tail)**

Reasoning:

- Sum of dependent indicators → approximate with Chernoff
- Bound probability of large deviation below mean

Q9: Inequality Proof

Goal:

$$\sum_{i=1}^{n} \sqrt{i^2+1} \geqslant \frac{n}{2}\sqrt{n^2+2n+5}$$

Approach:

- Use bounding techniques / averaging
- Recognize monotonic growth of terms

Q10: Poisson Tail Bounds

Given:

- $X \sim \text{Poisson}(20)$
- Want $P(X \geqslant 26)$

(a) Markov

$$P(X \geqslant 26) \leqslant \frac{20}{26}$$

(b) One-sided Chebyshev

$$P(X - 20 \geqslant 6) \leqslant \frac{\text{Var}(X)}{6^2} = \frac{20}{36}$$

(c) Chernoff Bound

Step 1: Use MGF

$$M_X(t) = e^{20(e^t-1)}$$

Step 2: Apply Chernoff

$$P(X \geqslant 26) \leqslant \inf_{t>0} \frac{M_X(t)}{e^{26t}}$$

Step 3: Optimize

Solve:

$$20e^t = 26 \Rightarrow e^t = 1.3$$

Substitute back to get tight bound.

Key Takeaways

1. When to use which inequality

- **Markov** → only mean known, upper tail
- **Chebyshev** → mean + variance, two-sided
- **Chernoff** → tighter bounds using exponentials
- **Jensen** → convexity arguments

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2. Strategy for solving problems

1. Identify random variable
2. Check available information (mean, variance)
3. Select inequality accordingly
4. Convert problem into standard form
5. Apply bound carefully

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3. Exam Tip

Always **justify why a specific inequality is used**:

- This is often **more important than the final number**

If you want, I can convert this into a **perfect LaTeX scribe file (ready to submit)** or **add diagrams/intuition for each inequality**.